Real analysis – new problems

Problems built on recent research results

Each problem is a special case of a published result or of a lemma used in its proof; the papers are in this folder. The problems are ordered by difficulty. The difficulty is the position of the problem in an IMC day: 1 easy, 2 below average, 3 medium, 4 above average, 5 hard.

Problem 1. *IMC level: 2/5 (below average)*

Let C be the middle-thirds Cantor set, i.e. the set of numbers $\sum_{k\geqslant 1} c_k 3^{-k}$ with all $c_k \in \{0,2\}$.

(a) Find all $t > 0$ for which there is an x with $x,\ x+t,\ x+2t \in C$.

(b) Find the largest possible length of an arithmetic progression with non-zero difference contained in C.

Based on: S. Sandberg-Clark, K. Taylor, A. Yavicoli, *Arithmetic-progression gap sets in Cantor sets*, arXiv:2608.14998, Proposition 2.1.

Problem 2. *IMC level: 2/5 (below average)*

Fix a positive integer k. Let $\varphi_k\colon \mathbb{N} \to \mathbb{N}$ be the bijection that exchanges every odd number m with $2^k m$ and fixes every n whose 2-adic valuation is neither 0 nor k. Prove that the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{\varphi_k(n)+1}}{\varphi_k(n)}$$

converges and find its sum.

Based on: M. Gilula, *A class of simple rearrangements of the alternating harmonic series*, arXiv:2107.01364, Theorem 1.

Problem 3. *IMC level: 2/5 (below average)*

Let $I \subseteq \{0,1,2,\dots\}$ and $I(k) = |I \cap \{0,1,\dots,k\}|$. Let A_I be the set of positive integers whose binary digit in position i (the coefficient of 2^i) is 0 for every $i \in I$. Prove that

$$\sum_{n\in A_I} \frac{1}{n} < \infty \iff \sum_{k\geqslant 0} 2^{-I(k)} < \infty.$$

Deduce that the series diverges for $I = \{1,2,4,8,\dots\}$ but converges for $I = \{1,4,9,16,\dots\}$.

Based on: M. B. Nathanson, *Curious convergent series of integers with missing digits*, arXiv:2006.13345, Lemma 1 and Theorem 1.

Problem 4. *IMC level: 2/5 (below average)*

Let $n \geqslant 1$ and let $x_1, \dots, x_n$ be pairwise distinct positive reals. Put $a_i = \prod_{j \neq i} \frac{x_j}{x_j - x_i}$ for $i = 1, \dots, n$ (the empty product is 1). Prove that

$$\frac{x_1 x_2 \cdots x_n}{n\,(1+x_1)\cdots(1+x_n)} < \sum_{i=1}^{n} a_i \ln(1+x_i) < \frac{x_1 x_2 \cdots x_n}{n}.$$

Based on: V. Bitsouni, N. Gialelis, D.-S. Marinescu, *An inequality for completely monotone functions*, arXiv:2204.06602, inequality (1) and Proposition 1.

Problem 5. *IMC level: 2/5 (below average)*

Let $p > 1$, let $f\colon (0,1] \to [0,\infty)$ be non-increasing and integrable, and put $F(x) = \int_0^x f(t)\,dt$. Prove that

$$\int_0^1 \Big(\frac{F(x)}{x}\Big)^p dx \;\geqslant\; \frac{p}{p-1} \int_0^1 (1 - x^{p-1}) f(x)^p \, dx,$$

and that equality holds for the indicator function of $(0, c]$ for every $c \in (0,1]$.

Based on: L.-E. Persson, N. Samko, G. Tephnadze, *Sharpness of some Hardy-type inequalities*, arXiv:2302.12298, Theorem 3.2(a) and Lemma 3.1.

Problem 6. *IMC level: 2/5 (below average)*

Let $f\colon [0,1] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Prove that there is a function $g\colon [0,1] \to \mathbb{R}$ with $|g(x) - g(y)| \leqslant \varepsilon |x - y|$ for all x, y such that for every $y \in \mathbb{R}$ the set $\{x \in [0,1] : f(x) + g(x) = y\}$ has Lebesgue measure zero.

Based on: S. Barza, M. Lind, *Eliminating positive-measure level sets by small Lipschitz perturbations*, arXiv:2607.02348, Theorem 1.1.

Problem 7. *IMC level: 2/5 (below average)*

Let $N \geqslant 1$ and let $p(x) = a_0 + \sum_{k=1}^{N} (a_k \cos kx + b_k \sin kx)$ be a real trigonometric polynomial, not identically zero, with $\int_{-\pi}^{\pi} p(x)\,dx \leqslant 0$. Suppose that $r \in (0, \pi]$ is such that $p(x) \geqslant 0$ for all x with $r \leqslant |x| \leqslant \pi$. Prove that $r \geqslant \frac{\pi}{N+1}$, and show that $r = \frac{\pi}{N+1}$ is possible.

Based on: T. Ismoilov, *Sharp sign uncertainty for trigonometric polynomials*, arXiv:2606.02299, Theorem 1.

Problem 8. *IMC level: 2/5 (below average)*

Let $f\colon [0,\pi] \to \mathbb{R}$ be continuously differentiable with $f(0) = f(\pi) = 0$. Prove that

$$\int_0^{\pi} f'(x)^2 \, dx \;\geqslant\; \frac{1}{4} \int_0^{\pi} f(x)^2 \Big(1 + \frac{1}{\sin^2 x}\Big) dx,$$

and show that neither of the two constants $\frac{1}{4}$ on the right-hand side (in front of 1 and in front of $\frac{1}{\sin^2 x}$) can be increased.

Based on: B. Rosenzweig, J. Stanfill, *An elementary approach to integral inequalities involving higher order derivatives*, arXiv:2509.14513, Example 4.7, formula (4.29).

Problem 9. *IMC level: 2/5 (below average)*

Let $f\colon [0,1] \to \mathbb{R}$ be convex. For $n \geqslant 1$ put $A_n = \frac{1}{n+1}\sum_{i=0}^{n} f\left(\frac{i}{n}\right)$, and for $n \geqslant 2$ put $B_n = \frac{1}{n-1}\sum_{i=1}^{n-1} f\left(\frac{i}{n}\right)$. Prove that the sequence (A_n) is non-increasing and the sequence (B_n) is non-decreasing. Deduce that $B_m \leqslant \int_0^1 f(x)\,dx \leqslant A_n$ for all $m \geqslant 2$ and $n \geqslant 1$.

Based on: J. Rooin, H. Dehghan, *Some monotonicity properties of convex functions with applications*, arXiv:1310.6698, Theorem 2.1 and Corollaries 2.3, 2.4.

Problem 10. *IMC level: 2/5 (below average)*

Let $f\colon [0,1] \to [0,\infty)$ be non-decreasing. Put $F_1(x) = \int_0^x f(t)\,dt$, $F_2(x) = \int_0^x F_1(t)\,dt$, $F_3(x) = \int_0^x F_2(t)\,dt$, and $a = F_1(1)$, $b = F_2(1)$, $c = F_3(1)$. Prove that $2b^2 \leqslant 3ac$.

Based on: M. R. Burke, M. Haris, Madhavendra, *Repeated integrals of increasing functions*, arXiv:2512.02151, Theorem 6.14.

Problem 11. *IMC level: 3/5 (medium)*

For a positive integer r let $P_r(x) = \sum_{k=0}^{r} \frac{(-x)^k}{k!} = 1 - x + \frac{x^2}{2!} - \cdots + \frac{(-x)^r}{r!}$. Find all r for which every coefficient of the Taylor series of $1/P_r(x)$ at $x = 0$ is non-negative.

Based on: J. Engbers, D. Galvin, C. Smyth, *Reciprocals of thinned exponential series*, arXiv:2303.14057, Introduction (Gessel's observation), Theorems 1.1 and 1.4.

Problem 12. *IMC level: 3/5 (medium)*

Let $(a_n)_{n\in\mathbb{Z}}$ be a strictly increasing sequence of real numbers such that for every n the series $\sum_{k\neq n}(a_k - a_n)^{-2}$ converges and

$$\sum_{k>n} \frac{1}{(a_k - a_n)^2} = \sum_{k<n} \frac{1}{(a_n - a_k)^2}.$$

Suppose that among the differences $g_n = a_{n+1} - a_n$ there is a largest one. Prove that all the differences g_n are equal.

Based on: A. Ferudun, *Equilibria of inverse-square repulsion on the line are arithmetic progressions*, arXiv:2610.06944, Lemma 2.3 and Remark 2.5.

Problem 13. *IMC level: 3/5 (medium)*

For a positive integer n let $s(n)$ be the number of ones in the binary expansion of n. For $k \geqslant 1$ put $A_k = \sum_{n\geqslant 1,\ s(n)=k} \frac{1}{n}$. Prove that every A_k is finite and that $\lim_{k\to\infty} A_k = 2\ln 2$.

Based on: J.-P. Allouche, C. Morin, *Kempner-like harmonic series*, arXiv:2305.18180, Theorem 1.

Problem 14. *IMC level: 3/5 (medium)*

Define $a_1 = 1$ and, for $n \geqslant 2$,

$$a_n = \begin{cases} a_{n-1} + 3, & \text{if } n \in \{a_1, \ldots, a_{n-1}\}, \\ a_{n-1} + 2, & \text{otherwise.} \end{cases}$$

Prove that $\lim\limits_{n\to\infty} \dfrac{a_n}{n} = 1 + \sqrt{2}$.

Based on: B. Cloitre, *A study of a family of self-referential sequences*, arXiv:2506.18103, Theorem 2.1 and Lemma 2.2.

Problem 15. *IMC level: 3/5 (medium)*

Let $f\colon [0,1] \to [0,1]$ be continuous, let $x_0 \in [0,1]$, and let

$$x_{n+1} = \frac{f(x_0) + f(x_1) + \cdots + f(x_n)}{n+1} \qquad (n \geqslant 0).$$

Prove that the sequence (x_n) converges.

Based on: R. May, *On the convergence of a perturbed one dimensional Mann's process*, arXiv:2504.16154, Theorem 1.2 (the theorem of Franks and Marzec).

Problem 16. *IMC level: 3/5 (medium)*

Let $D = \{(x,y) : x^2 + y^2 \leqslant 1\}$ and let $u\colon D \to \mathbb{R}$ be a concave Lipschitz function with $u = 0$ on the boundary of D (so u_x, u_y exist almost everywhere and are bounded).

(a) Prove that $\displaystyle\iint_D |u_x|\,dx\,dy \leqslant 2 \iint_D |u_y|\,dx\,dy$.

(b) Prove that the constant 2 cannot be replaced by a smaller number.

Based on: A. Plakhov, V. Protasov, *On inequalities between norms of partial derivatives on convex domains*, arXiv:2505.01611, Lemma 1, Theorems 1 and 2.

Problem 17. *IMC level: 3/5 (medium)*

Let $a_0 = 0$ and let $a_1, a_2, \ldots$ be real numbers, only finitely many of which are non-zero. Prove that

$$\sum_{n=1}^{\infty} (a_n - a_{n-1})^2 \;\geqslant\; \sum_{n=1}^{\infty} \Bigl(2 - \sqrt{1 - \tfrac{1}{n}} - \sqrt{1 + \tfrac{1}{n}}\Bigr) a_n^2,$$

and deduce that $\sum_{n\geqslant 1} (a_n - a_{n-1})^2 > \frac{1}{4} \sum_{n\geqslant 1} \frac{a_n^2}{n^2}$ unless all a_n are zero.

Based on: B. Das, A. Manna, *On improvements of the Hardy, Copson and Rellich inequalities*, arXiv:2309.04923, inequality (1.5) (Keller–Pinchover–Pogorzelski) and factorisation (1.6).

Problem 18. *IMC level: 4/5 (above average)*

Let $0 \leqslant r_0 < r_1 < r_2 < \cdots$ be real numbers such that $r_{j+1} - r_j \geqslant 1$ for all $j \geqslant 0$. Prove that for every $x > 0$

$$\sum_{j=0}^{\infty} \frac{x^{r_j}}{\Gamma(1+r_j)} \leqslant e^x.$$

Based on: G. Molteni, *A sharp Mittag-Leffler bound for series with separated exponents*, arXiv:2611.04540, Theorem and Lemma 1.

Problem 19. *IMC level: 4/5 (above average)*

Prove that there exists a continuously differentiable function $F\colon \mathbb{R}^2 \to \mathbb{R}$ such that the set of its critical values $\{F(p) : \nabla F(p) = 0\}$ contains the interval $[0, 2]$.

Based on: J. Ferrera, *An elementary example of Sard's theorem sharpness*, arXiv:2202.08000, Proposition 1 and Lemma 3.

Problem 20. *IMC level: 4/5 (above average)*

Let $(c_n)_{n\geqslant 1}$ be a non-decreasing sequence of positive reals with $c_n \to \infty$, and let $x = (0, c_1, 0, c_2, 0, c_3, \dots)$, i.e. $x_{2n-1} = 0$ and $x_{2n} = c_n$. Prove that there exists a bijection $p\colon \mathbb{N} \to \mathbb{N}$ such that

$$\lim_{m\to\infty} \frac{x_{p(1)} + x_{p(2)} + \cdots + x_{p(m)}}{m} = 1$$

if and only if $\lim\limits_{n\to\infty} \dfrac{c_n}{c_1 + c_2 + \cdots + c_{n-1}} = 0$.

Based on: A. Losonczi, *Points accessible in average by rearrangement of sequences*, arXiv:1901.01127, Introduction (Theorems 4.3 and 4.4 of part I) and Theorem 3.2.

Problem 21. *IMC level: 4/5 (above average)*

Let $1 \leqslant m \leqslant n$ be integers and let $f(x) = \sum_{k=m}^{n}(a_k \cos kx + b_k \sin kx)$ with real a_k, b_k, not all zero.

(a) Prove that the set $\{x \in [0, 2\pi) : f(x) > 0\}$ has Lebesgue measure at least $\frac{2\pi m}{m+n}$.

(b) Show that equality is possible when $n = m + 1$.

Based on: A. Ismailov, *Lower bounds on the measure of the support of positive and negative parts of trigonometric polynomials*, arXiv:2302.05308, Theorem 3.2 and Proposition 3.1.

Problem 22. *IMC level: 4/5 (above average)*

Let n be a positive integer and let $f\colon \mathbb{R} \to \mathbb{R}$ be a 2π-periodic function such that $|f(x)-f(y)| \leqslant |x-y|$ for all x, y and

$$\int_0^{2\pi} f(x)\cos kx\,dx = \int_0^{2\pi} f(x)\sin kx\,dx = 0 \qquad (k = 0, 1, \dots, n-1).$$

Prove that $|f(x)| \leqslant \frac{\pi}{2n}$ for all x.

Based on: P. Joharinad, J. Jost, S. Lim, R. Matveev, *Reverse Bernstein inequality on the circle*, arXiv:2302.10122, Theorem 1.A (the Bohr–Favard inequality).

Problem 23. *IMC level: 4/5 (above average)*

Let $a_1, \ldots, a_n$ be nonzero integers (not necessarily distinct). Let S be the set of $t \in [0,1]$ for which the numbers $\cos(2\pi a_1 t), \ldots, \cos(2\pi a_n t)$ are either all positive or all negative. Prove that the Lebesgue measure of S is at least 4^{1-n}.

Based on: C. Giannitsi, J. Madrid, *Exponential decay of threshold correlations for symmetric periodic functions*, arXiv:2610.38135, Theorem 1.2 and Proposition 3.1.

Problem 24. *IMC level: 4/5 (above average)*

Let $f\colon \mathbb{R} \to \mathbb{R}$ satisfy $|f(x) - f(y)| \leqslant |x - y|$ for all x, y, and $f(x) = 0$ for $|x| \geqslant 1$. For $x \in \mathbb{R}$ let

$$Mf(x) = \sup\Big\{ \frac{1}{b-a} \int_a^b |f(t)|\, dt : \ a \leqslant x \leqslant b,\ a < b \Big\}.$$

Prove that $|Mf(x) - Mf(y)| \leqslant (\sqrt{2}-1)|x-y|$ for all $x, y \in \mathbb{R}$, and that the constant $\sqrt{2}-1$ cannot be replaced by a smaller one.

Based on: S.-Y. Liao, J. Madrid, E. A. Palsson, *Sharp derivative inequalities for maximal functions*, arXiv:2610.39178, Introduction (the inequality of Aldaz, Colzani and Pérez Lázaro).

Problem 25. *IMC level: 4/5 (above average)*

Let T be the set of sums of all subseries of the series

$$\frac{3}{4} + \frac{2}{4} + \frac{3}{4^2} + \frac{2}{4^2} + \frac{3}{4^3} + \frac{2}{4^3} + \cdots,$$

that is, $T = \{\sum_{i \geqslant 1} d_i 4^{-i} : \ d_i \in \{0,2,3,5\}\}$.

(a) Prove that T contains the interval $\left[\frac{2}{3}, 1\right]$.

(b) Prove that T is not a finite union of closed intervals and points.

Based on: M. Pratsiovytyi, D. Karvatskyi, *Fractal analysis of Guthrie–Nymann's set and its generalisations*, arXiv:2405.16576, Section 2.